A Certified Nested-Grid Continuation of a Physical Spectral Count

Dyadic Schur Self-Energy and a 2048-to-4096 Dimension Transfer
at a Quadratic Band-Merging Map

Liang Wang

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Abstract

A preceding finite-model certificate proved that one exact stored 2048-dimensional Perron/parity-extracted physical two-step matrix has exactly one eigenvalue inside a prescribed complex circle. That theorem did not show that the count survives grid refinement. We give the first rigorous adjacent-dimension continuation for this stored model, at fixed Gaussian noise width $\sigma = 10^{-2}$, from 2048 to 4096 midpoint cells.

The fine grid is decomposed by exact dyadic replication and alternating-detail coordinates. Four rank-96 stored approximants, certified by componentwise factor-graph residuals, bound the coarse consistency, two cross channels, and detail block by 1.98878×10^{-4}, 1.13215×10^{-2}, 1.37227×10^{-2}, and 1.30523×10^{-4}, respectively. The detail spectrum is excluded from the counting disk, and its Schur self-energy is at most 4.13077×10^{-4}. Thus the fine effective coarse block differs from the stored coarse physical matrix by at most 6.11954×10^{-4} on the full contour.

A sparse-Grushin atlas with 170 rigorous centers and an exact 314-leaf rational partition bounds the matrix Rouché product by $0.899350 < 1$. A bitwise replay identifies the coarse object with the preceding one-count certificate. Consequently, for the exact stored binary64 matrices,

$$\boxed{N_\Gamma(A_{2048}) = N_\Gamma(A_{4096}) = 1.}$$

This is a finite stored-matrix theorem at one fixed noise scale. It is not a continuum, arbitrary-dimension, or zero-noise result.

1 Introduction

Finite-dimensional spectral discoveries become mathematically useful only after two logically separate questions are answered. First, does a specified stored matrix really have the reported contour count? Second, does that count survive a change of discretization? The first question was closed for the 2048-dimensional physical two-step matrix in Wang [10]. The second remained open.

The distinction is substantial for nonnormal transfer matrices. A visually stable eigenvalue and a small entrywise grid change do not imply a stable spectral count; the relevant amplification is controlled by the resolvent [6, 8]. Moreover, matrices at different dimensions do not live in the same coordinate space. A subtraction becomes meaningful only after a lifting, restriction, and detail space are specified.

This paper fixes the Gaussian width at

$$\sigma = 10^{-2}$$

and compares the exact stored midpoint matrices at dimensions

$$n = 2048, \qquad 2n = 4096.$$

The refinement is dyadic, so the fine space admits an exact binary coarse/detail splitting. The main mechanism is

$$\text{fine matrix} \longrightarrow \text{detail Schur complement} \longrightarrow \text{small analytic self-energy} \\ \longrightarrow \text{coarse resolvent Rouché comparison.}$$

Three technical points make the argument rigorous.

1. The coordinate transform uses only copying, sign changes, and division by two. No interpolation constant or irrational normalization enters the exact algebra.
2. The four cross-grid blocks are not bounded by their Frobenius norms. Stored rank-96 singular centers are combined with componentwise residual enclosures. The centers are proof generators; the final inequalities include their full residuals.
3. The coarse physical resolvent is certified directly. A sparse two-step Grushin realization supplies approximate inverses, while exact stored-factor residuals convert them into rigorous bounds.

The resulting theorem is a genuine discretization-stability step, but a limited one. It compares two finite stored matrices at one noise width. It does not establish a uniform $n \to \infty$ estimate, despite the general collectively compact framework for smooth integral kernels [2]. It also does not move the noise parameter.

2 Stored physical matrices and evidence levels

Let P_m denote the stored row-normalized folded Gaussian midpoint matrix on m positive cells. The deterministic map parameter, Gaussian cutoff, row normalization, sparse indices, and all nonzero weights are archived in the machine-readable snapshot. Let

$$X_m,\ Y_m \in \mathbb{R}^{m\times 2}, \qquad \Lambda_m \in \mathbb{R}^{2\times 2}$$

be the stored right modes, left modes, and peripheral eigenvalue diagonal. The Perron/parity-extracted one-step and physical two-step matrices are

$$U_m = P_m - X_m \Lambda_m Y_m^T, \qquad A_m = U_m^2. \tag{1}$$

Every binary64 number in these factors is treated as an exact real number. The theorem concerns

$$A_{\mathrm{c}} = A_{2048}, \qquad A_{\mathrm{f}} = A_{4096}.$$

The counting contour is the positively oriented circle

$$\Gamma: \quad z = z_0 + re^{i\theta}, \qquad 0 \le \theta \le 2\pi, \tag{2}$$

with stored parameters

$$z_0 = -0.3233504401504541 - 0.5508412474453575i, \qquad r = 0.2624987592858511. \tag{3}$$

The distance from the origin to the closed counting disk is bounded below by

$$d_0 = |z_0| - r \ge 0.376235604149098. \tag{4}$$

We distinguish three evidence levels.

Exact algebra

Coordinate identities and determinant factorizations over the exact stored real numbers.

Rigorous certificate

Outward bounds for the exact stored binary64 target under the stated round-to-nearest model, with Arb used for final scalar and circular-arc comparisons [5, 7].

Floating diagnostic

Sparse eigensolves and singular vectors used to choose proof objects and to display locations. They do not establish the theorem.

3 Exact dyadic coarse/detail coordinates

For $x, y \in \mathbb{C}^n$, define $J, K : \mathbb{C}^n \to \mathbb{C}^{2n}$ by

$$(Jx)_{2j} = (Jx)_{2j+1} = x_j, \tag{5}$$

$$(Ky)_{2j} = y_j, \qquad (Ky)_{2j+1} = -y_j. \tag{6}$$

Define

$$R = \tfrac{1}{2}J^T, \qquad S = \tfrac{1}{2}K^T. \tag{7}$$

These are unnormalized Haar coarse/detail coordinates. Their advantage is exact representability in binary arithmetic.

Lemma 3.1 (Exact dyadic inverse)**.** *The maps satisfy*

$$RJ = SK = I_n, \qquad RK = SJ = 0,$$

and therefore

$$T = [J \ K], \qquad T^{-1} = \begin{pmatrix} R \\ S \end{pmatrix}. \tag{8}$$

Proof. Every identity follows independently on each fine-grid pair. The sum of the two replicated entries is $2x_j$, while the difference of the alternating entries is $2y_j$. The cross sum and cross difference vanish. □

In these coordinates write

$$T^{-1}A_{\mathrm{f}}T = \begin{pmatrix} A_{\mathrm{cc}} & B_{\mathrm{cd}} \\ C_{\mathrm{dc}} & D_{\mathrm{dd}} \end{pmatrix}, \tag{9}$$

where

$$A_{\mathrm{cc}} = RA_{\mathrm{f}}J, \quad B_{\mathrm{cd}} = RA_{\mathrm{f}}K, \quad C_{\mathrm{dc}} = SA_{\mathrm{f}}J, \quad D_{\mathrm{dd}} = SA_{\mathrm{f}}K. \tag{10}$$

The coarse consistency defect is

$$E_{\mathrm{cc}} = A_{\mathrm{cc}} - A_{\mathrm{c}}. \tag{11}$$

Proposition 3.2 (Exact Schur determinant identity)**.** *If $zI - D_{\mathrm{dd}}$ is invertible, then*

$$\det(zI_{2n} - A_{\mathrm{f}}) = \det(zI_n - D_{\mathrm{dd}}) \det \mathcal{F}_{\mathrm{f}}(z), \tag{12}$$

$$\mathcal{F}_{\mathrm{f}}(z) = zI_n - A_{\mathrm{c}} - \Delta(z), \tag{13}$$

$$\Delta(z) = E_{\mathrm{cc}} + B_{\mathrm{cd}}(zI_n - D_{\mathrm{dd}})^{-1}C_{\mathrm{dc}}. \tag{14}$$

Proof. Similarity by T preserves the characteristic determinant. Taking the Schur complement of the lower-right block in $zI - T^{-1}A_{\mathrm{f}}T$ gives (12). Substituting $A_{\mathrm{cc}} = A_{\mathrm{c}} + E_{\mathrm{cc}}$ gives (13)–(14). □

Table 1: Rigorous block bounds for the exact stored nested-grid pair. The residual column is the Frobenius norm after subtracting the stored rank-96 center.

block	leading stored scale	residual upper	spectral-norm upper
E_{cc}	$1.98874394 \times 10^{-4}$	2.78257×10^{-9}	$1.98877177 \times 10^{-4}$
C_{dc}	$1.13214077 \times 10^{-2}$	3.02114×10^{-8}	$1.13214379 \times 10^{-2}$
B_{cd}	$1.37226274 \times 10^{-2}$	1.42285×10^{-8}	$1.37226417 \times 10^{-2}$
D_{dd}	$1.30520845 \times 10^{-4}$	1.23210×10^{-9}	$1.30522077 \times 10^{-4}$

4 Certified low-rank bounds for the four blocks

Direct Frobenius estimates are insufficient. The floating Frobenius candidates for the two cross blocks are approximately 4.23×10^{-2} and 2.45×10^{-2}, whereas their spectral norms are about 1.13×10^{-2} and 1.37×10^{-2}. This difference decides whether the final Rouché gate closes.

For each exact stored block E, a floating sparse singular calculation produces stored matrices

$$L \in \mathbb{R}^{n \times k}, \qquad \Sigma = \operatorname{diag}(s_1, \ldots, s_k), \qquad V^T \in \mathbb{R}^{k \times n},$$

with $k = 96$. These arrays are not trusted as singular vectors. They define only an exact stored low-rank comparison matrix $L\Sigma V^T$.

Lemma 4.1 (Low-rank residual bound)**.** *Suppose componentwise arithmetic proves*

$$\left\|L^T L - I\right\|_F \le \delta_L, \qquad \left\|V V^T - I\right\|_F \le \delta_V, \qquad \left\|E - L\Sigma V^T\right\|_F \le \rho.$$

Then

$$\|E\|_2 \le \sqrt{1+\delta_L}\, \max_j |s_j|\, \sqrt{1+\delta_V} + \rho. \tag{15}$$

Proof. The Gram bounds imply $\|L\|_2^2 \le 1+\delta_L$ and $\left\|V^T\right\|_2^2 \le 1+\delta_V$. Apply submultiplicativity to the stored low-rank center and use the Frobenius norm to bound the residual spectral norm. □

The residual is evaluated in blocks of 128 basis columns. Every action of P_m, $X_m \Lambda_m Y_m^T$, the two-step composition, and the dyadic coordinate maps propagates one outward radius per output entry. The block Frobenius bounds are combined by an outward Euclidean sum [4, 9].

The left and right Gram-defect Frobenius bounds are all below 6.10×10^{-11}. Thus the stored low-rank factors are close to orthonormal, but the certificate uses the displayed defect bounds rather than setting them to zero.

5 Detail exclusion and the effective perturbation gate

The detail block is concentrated near the origin, whereas the counting disk is separated from the origin by (4).

Proposition 5.1 (No detail eigenvalue in the counting disk)**.** *For the exact stored detail block,*

$$N_\Gamma(D_{\mathrm{dd}}) = 0.$$

Moreover, on Γ*,*

$$\left\|(zI - D_{\mathrm{dd}})^{-1}\right\|_2 \le 2.658831394912469. \tag{16}$$

Proof. By table 1, every detail eigenvalue lies in $|\lambda| \le 1.30522077 \times 10^{-4}$. This disk is disjoint from the closed counting disk because its radius is smaller than d_0. Hence the detail count is zero. On the boundary,

$$\left\|(zI - D_{\mathrm{dd}})^{-1}\right\|_2 \le \frac{1}{|z| - \|D_{\mathrm{dd}}\|_2} \le \frac{1}{d_0 - \|D_{\mathrm{dd}}\|_2},$$

which gives (16) after outward rounding. □

Combining table 1 and proposition 5.1 gives

$$\sup_{z\in\Gamma} \left\|B_{\mathrm{cd}}(zI - D_{\mathrm{dd}})^{-1}C_{\mathrm{dc}}\right\|_2 \le 4.130761388556052 \times 10^{-4}. \tag{17}$$

Therefore

$$\boxed{\sup_{z\in\Gamma} \|\Delta(z)\|_2 \le \varepsilon := 6.119533154024968 \times 10^{-4}.} \tag{18}$$

The corresponding admissible coarse resolvent threshold is

$$\varepsilon^{-1} \ge 1634.1115814402854. \tag{19}$$

6 Bitwise coarse-count replay

The coarse count cannot simply be quoted for a vaguely similar reconstruction. The RH-36 snapshot stores the coarse sparse matrix, peripheral factors, packet pair, and every low-rank proof center. We replay the exact packet-pair correction of Wang [10] against these stored coarse arrays.

The replay recomputes the exact dyadic packet Gram defect, physical block majorants, and all 949 inherited complement/Feshbach leaf transfers. Two independent archives match bitwise:

$$\begin{aligned} \mathrm{SHA256}(\text{exact pair defect}) &= \texttt{a1c2b4561c5f...aa7393e}, \\ \mathrm{SHA256}(\text{949-leaf transfer ledger}) &= \texttt{97c5d033e7a6...84242eff}. \end{aligned}$$

The maximum replayed complement product is 3.76781×10^{-9}, and the maximum replayed Feshbach Rouché product is 0.547185. Consequently, the exact coarse object in the present snapshot is the object certified previously, and

$$N_\Gamma(A_{\mathrm{c}}) = 1. \tag{20}$$

7 A direct coarse physical resolvent atlas

The remaining requirement is a full-boundary bound for the coarse physical resolvent, below the threshold in (19). A packet/complement block norm estimate is unsuitable: the complement resolvent can be very large, while the physical inverse contains essential low-rank cancellation. We therefore certify the physical inverse directly.

7.1 Sparse two-step realization

The formal two-step linearization

$$\mathcal{G}_0(z) = \begin{pmatrix} zI & -U_{2048} \\ -U_{2048} & I \end{pmatrix} \tag{21}$$

has Schur complement $zI - A_{\mathrm{c}}$. Since U_{2048} is a sparse matrix plus two peripheral channels, four bordered channels represent (21) without forming a dense correction. Sparse LU is used only to generate an approximate right inverse R_0. The exact stored physical action is then reevaluated columnwise with componentwise radii. If

$$\rho_0 = \|I - (\zeta I - A_{\mathrm{c}})R_0\|_F < 1, \tag{22}$$

then

$$\left\|(\zeta I - A_{\mathrm{c}})^{-1}\right\|_2 \leq \frac{\|R_0\|_F}{1-\rho_0}. \tag{23}$$

This is a standard a posteriori Neumann argument [4, 7]. The maximum certified center residual is 4.516×10^{-9}.

7.2 Arc transport

At a certified center ζ with inverse bound M_ζ, the resolvent identity gives

$$\left\|(zI - A_{\mathrm{c}})^{-1}\right\|_2 \leq \frac{M_\zeta}{1 - M_\zeta|z-\zeta|} \tag{24}$$

whenever the denominator is positive. Circular subarcs are enclosed by Arb-backed complex discs whose endpoints are exact rational turns. Adaptive bisection produces an exact partition of the full circle.

The final atlas contains 170 rigorous centers. Center bounds range from 50.7898 to 703.401. An exact 314-leaf rational partition has no unresolved leaf. The largest transported bound is

$$\sup_{z\in\Gamma} \left\|(zI - A_{\mathrm{c}})^{-1}\right\|_2 \leq 1469.637830624761 \tag{25}$$

in the leafwise sense recorded by the atlas. The archive deliberately targets a 0.9 continuation product rather than stopping at the theorem threshold.

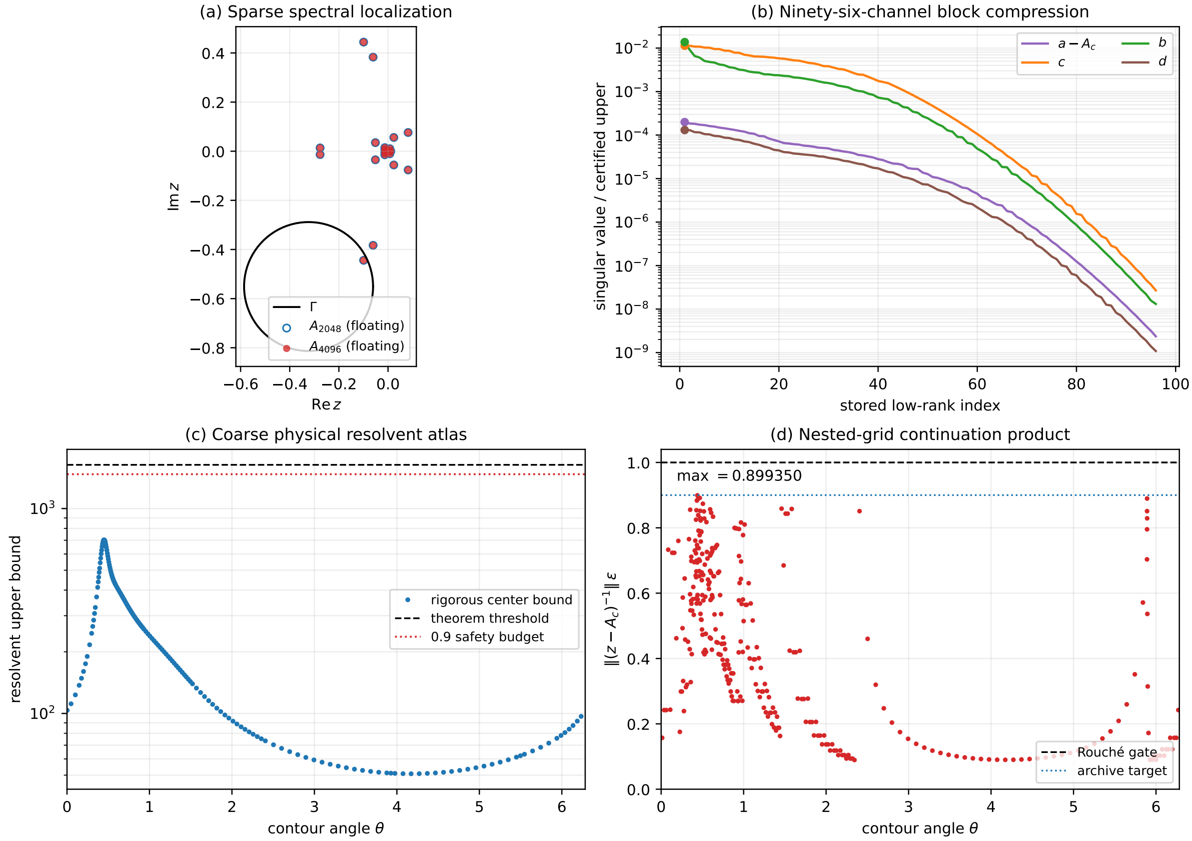

Figure 1: Nested-grid certificate. (a) Floating sparse eigenvalues are shown only for localization. (b) The four stored rank-96 block centers have rapidly decaying singular values; the first markers show rigorous final norm uppers. (c) Direct physical center-resolvent bounds remain below the theorem threshold. (d) Every rational leaf satisfies the matrix Rouché gate, with maximum 0.899350.

8 The nested-grid count theorem

Theorem 8.1 (Certified 2048-to-4096 count continuation). *For the exact stored binary64 matrices* (1) *at* $\sigma = 10^{-2}$, *on the contour* (3),

$$\boxed{N_\Gamma(A_{2048}) = N_\Gamma(A_{4096}) = 1.} \tag{26}$$

Both counts include algebraic multiplicity.

Proof. The coarse count is one by the bitwise replay (20). By proposition 5.1, the detail resolvent is analytic on and inside Γ, and its interior count is zero. Hence the fine determinant has the exact factorization (12) throughout the counting domain.

For the effective factor, (13) gives

$$\mathcal{F}_f(z) = (zI - A_c) - \Delta(z).$$

On every one of the 314 rational leaves, the certified atlas and (18) give

$$\left\|(zI - A_c)^{-1}\Delta(z)\right\|_2 \le 0.8993497428917556 < 1. \tag{27}$$

The matrix Rouché theorem therefore preserves the determinant winding and zero count between $zI - A_c$ and $\mathcal{F}_f(z)$ [1, 3]. Thus

$$N_\Gamma(\mathcal{F}_f) = N_\Gamma(A_c) = 1.$$

Adding the zero detail count through (12) yields $N_\Gamma(A_f) = 1$. □

Remark 8.2 (What the theorem does not use). The proof does not compare two floating eigenvalue lists, invoke a condition number of an eigenvector matrix, or assume that a rank-96 approximation is exact. Floating arithmetic chooses proof centers. Every final gate is an outward inequality for the exact stored matrices.

9 Floating localization and reproducibility

A 24-mode sparse eigensolve is archived as a diagnostic. It resolves one lower-half-plane eigenvalue inside the contour at each dimension:

$$\begin{aligned} \lambda_{2048}^{\mathrm{fl}} &= -0.0993604146537118 - 0.4442017426458030i, \\ \lambda_{4096}^{\mathrm{fl}} &= -0.0994035212675019 - 0.4441920040123497i. \end{aligned} \tag{28}$$

Their floating displacement is

$$|\lambda_{4096}^{\mathrm{fl}} - \lambda_{2048}^{\mathrm{fl}}| = 4.41930 \times 10^{-5}.$$

The maximum reported eigenpair residual is below 10^{-15}. These values are consistent with theorem 8.1, but they are not validated eigenvalue disks and do not enter its proof.

The formal archive contains:

1. a 35 MiB compressed snapshot of both exact sparse targets and the four stored rank-96 proof centers;
2. componentwise low-rank residual certificates and hashes;
3. 170 center certificates consolidated in a CSV ledger;
4. an exact 314-leaf rational contour ledger;
5. the bitwise coarse-count replay and its 949-leaf inherited ledger;
6. the final theorem JSON, dependency manifest, tests, and figure data.

All center factorizations are replaceable. Their sparse LU factors are not trusted or archived as exact objects. Only the approximate inverse columns, their exact-target residual hashes, and the resulting outward bounds define the proof record.

10 Scope and next gates

The theorem establishes one finite-scale stability step:

$$2048 \longrightarrow 4096 \qquad (\sigma = 10^{-2}).$$

It does not yet establish any of the following.

1. No bound relates either stored matrix to the exact continuum Gaussian integral operator. General Nyström convergence theory does not supply the explicit nonnormal contour constants used here automatically [2].
2. The block bounds are certified only for one adjacent pair. A uniform estimate in n, or an induction over all dyadic refinements, remains open.
3. The noise width is fixed. Continuation to $\sigma = 4 \times 10^{-3}$ changes both dimension and packet geometry and requires a new resolvent atlas.

4. The unique fine eigenvalue is counted but not enclosed in a validated complex disk.
5. No self-adjoint generator, prime-power trace formula, zeta-zero identification, Hilbert–Pólya construction, or implication for the Riemann hypothesis is obtained.

The next direct experiment is the dyadic step $4096 \to 8192$ at the same noise width. If the coarse/detail norms continue to decrease while the physical resolvent remains controlled, the method can begin an empirical uniformity study. If a gate grows, the failing inequality identifies the obstruction. A stronger route would derive analytic quadrature bounds for the four Haar-coordinate blocks and combine them with a validated continuum resolvent estimate.

11 Conclusion

The finite physical count at dimension 2048 is not an isolated numerical accident under the first dyadic refinement. Exact coarse/detail coordinates reduce the 4096-dimensional problem to a detail factor and an analytic self-energy acting on the coarse space. Componentwise rank-96 residual certificates bound the total effective perturbation by 6.11954×10^{-4}. A direct physical resolvent atlas then closes the full-contour Rouché comparison with product 0.899350.

The resulting statement is precise and deliberately finite:

$$N_\Gamma(A_{2048}) = N_\Gamma(A_{4096}) = 1$$

for two exact stored binary64 matrices at $\sigma = 10^{-2}$. This supplies a rigorous first bridge from a one-grid physical theorem toward discretization stability, while leaving continuum and small-noise limits as separate gates.

A Coordinate formulas used by the certificate

For a fine vector $v \in \mathbb{C}^{2n}$, the implementation evaluates

$$(Rv)_j = \tfrac{1}{2}(v_{2j} + v_{2j+1}), \qquad (Sv)_j = \tfrac{1}{2}(v_{2j} - v_{2j+1}).$$

The exact block actions are

$$\begin{aligned} E_{\mathrm{cc}}x &= RA_{\mathrm{f}}Jx - A_{\mathrm{c}}x, & C_{\mathrm{dc}}x &= SA_{\mathrm{f}}Jx, \\ B_{\mathrm{cd}}y &= RA_{\mathrm{f}}Ky, & D_{\mathrm{dd}}y &= SA_{\mathrm{f}}Ky. \end{aligned}$$

Their Euclidean adjoints are evaluated by reversing the maps and applying the stored adjoint physical actions. Unit tests compare all four actions and adjoints with dense synthetic formulas and independently test the Schur determinant identity.

B Outward arithmetic assumptions

The componentwise graph assumes IEEE binary64 round-to-nearest arithmetic, finite intermediate values, no overflow, and no harmful underflow. Sparse and dense dot-product errors use conservative operation-count factors in the standard model of Higham [4]. Positive scalar combinations are rounded one binary64 step outward; circular geometry and final comparisons use 256-bit Arb. Exact packet Gram entries are dyadic rationals computed with integer arithmetic. These assumptions and the stored object hashes are recorded in the dependency manifest.

C Reproduction commands

From the paper directory, the fast archive checks are

```
PYTHONDONTWRITEBYTECODE=1 python -m pytest -q -p no:cacheprovider
PYTHONDONTWRITEBYTECODE=1 python experiments/build_archive.py
MPLBACKEND=Agg python experiments/make_figures.py
python experiments/verify_archive.py
```

The 170 center certificates are resumable. A fresh initial batch is generated by

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
python experiments/run_physical_resolvent_batch.py \
  --initial-grid 64 --workers 8 --chunk-size 256
```

and unresolved midpoint targets are supplied by the atlas JSON until the exact partition closes.

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